



**PES University, Bengaluru**  
(Established under Karnataka Act 16 of 2013)

**END SEMESTER ASSESSMENT (ESA) - Dec 2024**

**UE21CS351B - Cloud Computing**

**Total Marks : 100.0**

1.a. A company named "TechSolutions" specializes in banking software. They need high levels of data security and compliance with government regulations but want to reduce their infrastructure costs. Which cloud deployment model TechSolutions should use and why? (4.0 Marks)

1.b. Explain the generic cloud architecture with diagram (6.0 Marks)

1.c. What is SaaS? Explain functions of SaaS and list few examples (5.0 Marks)

1.d. Differentiate between microservice and monolithic model. Why the migration is not easy? (5.0 Marks)

2.a. What are the essential characteristics for VMM and instructions according to popek and goldberg principles? (6.0 Marks)

2.b. Bring out the comparison between hot migration and cold migration? Explain the different copy techniques of hot migration. (8.0 Marks)

2.c. What are controller-manager, kubelets and pods in Kubernetes? Explain with a diagram where each of them execute on master or worker? (6.0 Marks)

3.a. What constitutes cloud storage? Explain logical volume manager with diagram. (10.0 Marks)

3.b. Discuss the pitfalls in leaderless replication and approaches to address the challenges. (10.0 Marks)

4.a. Explain Bully algorithm with analysis.

(8.0 Marks)

4.b. What is fault tolerance? What are the considerations for building fault tolerance? (6.0 Marks)

4.c. What is apache zookeeper? Why it is required? List the objectives of zookeeper. (6.0 Marks)

5.a. Explain the differences between forward proxy and reverse proxy  
(4.0 Marks)

5.b. What is Cloud security? Explain the legal issues in cloud security  
(4.0 Marks)

5.c. What is Cloud Bursting? What is the need for cloud Bursting?  
(6.0 Marks)

5.d. What is DoS ? Explain the differences between DDoS and EDoS  
(6.0 Marks)